A guanine-nucleotide checkpoint for the type-2 T-cell program: perturb-seq nominates mycophenolic acid for eosinophilic asthma

Genome-scale $CD4^{+}$ T-cell CRISPRi screen → druggable metabolic regulator → natural-product therapeutic

Computational target-discovery report | Analysis of Marson-lab genome-scale Perturb-seq (bioRxiv 2025, doi:10.64898/2025.12.23.696273) | Data: CZI Virtual Cell Models · GEO GSE314342 · SRA SRP643211

Abstract. Genome-wide association studies (GWAS) implicate T-cell programs in autoimmune and allergic disease but rarely name a druggable node. In a genome-scale CRISPR-interference Perturb-seq screen of primary human $CD4^{+}$ T cells, each gene knockdown is a genetic mimic of inhibiting that gene's product, so a knockdown that selectively silences a pathogenic cytokine program marks a therapeutic target. Streaming only cytokine-readout columns from the study's 16.8 GB differential-expression matrix, we screened 11,526 knockdowns for selective suppression of effector programs. **Inosine-5′-monophosphate dehydrogenase 2 (IMPDH2)**, the rate-limiting enzyme of de novo guanine-nucleotide synthesis, ranked first of 576 druggable metabolic enzymes: its knockdown collapsed the type-2 cytokines IL-4/IL-5/IL-13 while sparing Th1, regulatory, and proliferative readouts. IMPDH2 is the target of **mycophenolic acid**, a natural Penicillium metabolite. We predict mycophenolic acid suppresses type-2 (eosinophilic) asthma, and propose a first experiment on allergen-stimulated patient $CD4^{+}$ T cells.

Keywords: Perturb-seq · $CD4^{+}$ T cells · IMPDH2 · mycophenolic acid · type-2 inflammation · eosinophilic asthma · IL-5 · drug repurposing

1 Background and rationale

Autoimmune and allergic diseases are driven by $CD4^{+}$ T helper cells that differentiate into effector states — Th1 (IFN-γ), Th17 (IL-17/IL-22), and type-2/Th2 (IL-4/IL-5/IL-13) — each defining a distinct disease axis. Human genetics has catalogued hundreds of risk genes for these conditions, but most encode transcription factors, receptors, or non-coding variants that are difficult to drug directly. The therapeutic question is not only which genes carry risk, but which druggable nodes control the pathogenic program downstream of that risk.

The genome-scale Perturb-seq atlas of Marson and colleagues provides an experimental route to that question. The authors used CRISPRi to knock down 12,748 expressed genes with 26,504 guides across ~22 million primary human $CD4^{+}$ T cells from four donors, profiling single-cell transcriptomes at rest and after 8 h / 48 h of TCR stimulation. Because CRISPRi produces a graded loss of function, each knockdown approximates pharmacological inhibition of the corresponding protein. This licenses a direct strategy: **scan every knockdown for one that switches off a pathogenic effector-cytokine program, then ask whether a natural small molecule inhibits that target.** We did not pre-commit to a disease; we let the data nominate both the actionable target and the effector axis it controls, then matched that axis to the condition whose genetics it best explains. The screen converged on a metabolic enzyme controlling the type-2 program — the axis dominating the genetics of eosinophilic asthma — and on a natural product with an established but never genome-nominated link to it.

2 Methods — how the dataset was used

Tractable access to a 1.8 TB resource. The raw atlas (per-cell matrices totalling ~1.8 TB) is far too large for the available hardware and was never downloaded. We used two processed layers: the authors' supplementary tables (differential-expression summaries, functional clusters, druggability and Open Targets GWAS annotations) from the project GitHub, and the consolidated differential-expression object `GWCD4i.DE_stats.h5ad` (16.8 GB) on the CZI public S3 bucket. Rather than pull the whole object, we opened it over the network and used HDF5 partial reads to stream only the 96 cytokine/effector readout columns across all 33,983 perturbation×condition contrasts — a <30 MB working matrix at <1 GB peak memory.

Effect definition and screen. Each contrast is a DESeq2 pseudobulk comparison of a knockdown against non-targeting controls, per condition; a negative $\log_2$ fold-change ($\log_2$FC) on an effector cytokine means knockdown suppresses it, mimicking target inhibition. We validated this sign convention with the positive control MEN1, recovering its reported role as a strong negative regulator of IL-10 (knockdown raises IL-10, $\log_2$FC +4.4, adjusted p = 2×10^{-5}). Readouts were grouped into type-2 (IL-4/IL-5/IL-13), Th1 (IFN-γ/TNF/LT-α), Th17 (IL-17F/IL-22/IL-23R/IL-6/CSF2), regulatory (FOXP3/CTLA4/IL-2RA/IL-10) and proliferation (IL-2) modules. For each knockdown with significant on-target activity we scored suppression of each effector program and its selectivity (sparing of the regulatory and proliferative arms), restricting candidates to druggable classes (enzymes, transporters, kinases, GPCRs, nuclear receptors, ion channels) annotated with Open Targets autoimmune-GWAS scores. Robustness used cross-donor/guide reproducibility; pathway enrichment used a one-sided Mann–Whitney U test. Analyses used Python (h5py, anndata, numpy, pandas, scipy, matplotlib).

3 Results

3.1 A single druggable enzyme selectively silences the type-2 program

Of the 11,526 gene knockdowns screened, 7,155 showed confirmed on-target activity at 8 h stimulation; ranking these by type-2 effect, the strongest suppressors were core TCR-signalling genes (VAV1, CD3E/G, LCP2, LAT) — essential and not selectively druggable. Filtering to druggable metabolic enzymes and transporters, **IMPDH2 ranked first of 576** (Table 1) and 16th of 7,155 genome-wide — the top 0.2%, 4.0 SD below the genome-wide mean. The effect was selective: IMPDH2 knockdown drove the type-2 module down by $\log_2$FC −4.5 while leaving Th1 essentially unchanged (−0.3), sparing the regulatory arm (−0.5), and slightly increasing IL-2 (Fig. 1, Fig. 2).

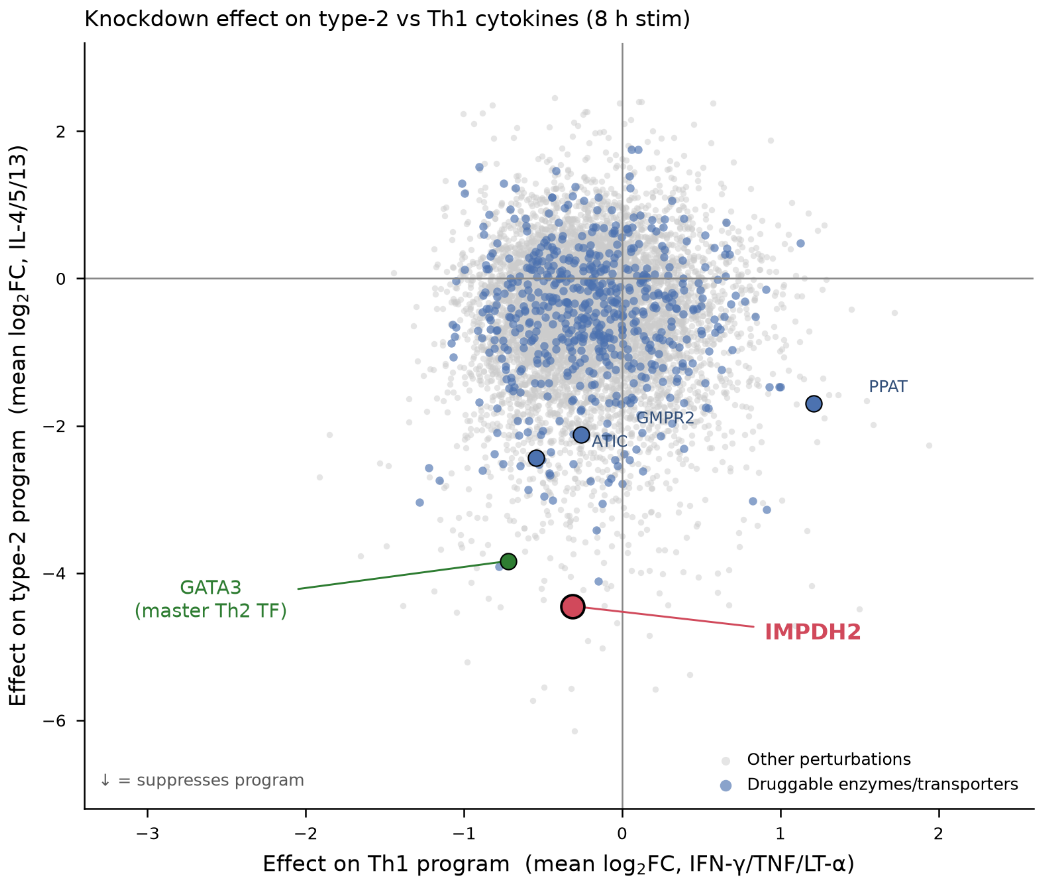

Figure 1. Genome-scale selectivity landscape. Each point is one gene knockdown (8 h stimulation, significant on-target knockdowns; n = 7,155), plotted by its effect on the Th1 program (x) versus the type-2 program (y); lower = stronger suppression. Druggable enzymes/transporters are highlighted (blue). IMPDH2 (red) sits at the extreme of type-2 suppression while remaining Th1-neutral, adjacent to GATA3 (the master type-2 transcription factor) and clustered with de novo purine-pathway enzymes (ATIC, PPAT, GMPR2). Other perturbations in grey.

Table 1. Top druggable metabolic knockdowns ranked by type-2 program suppression (8 h stimulation). $\log_2 FC < 0$ = knockdown suppresses the program. IMPDH2 combines the strongest type-2 suppression with Th1/regulatory sparing, a positive IL-2 (proliferation) readout, a large downstream footprint, and high cross-donor reproducibility.

Gene	Class	type-2 log₂FC	Th1 log₂FC	Treg log₂FC	IL-2 log₂FC	n downstream	cross-donor r
IMPDH2	enzyme	**−4.45**	−0.31	−0.46	+1.07	526	0.82
MAP3K5	enzyme	**−4.11**	−0.15	−1.38	−0.91	198	0.80
METAP2	enzyme	**−3.91**	−0.78	−2.04	−0.11	375	0.76
ATP6V0A2	transporter	**−3.42**	−0.16	−0.75	+2.36	39	0.43
MAP2K1	enzyme	**−3.13**	+0.91	−0.14	+0.03	2	—
CERS5	enzyme	**−3.04**	−1.28	−0.54	−0.26	0	—

3.2 The knockdown phenocopies loss of the master type-2 transcription factor

At the individual-cytokine level (Table 2), IMPDH2 knockdown at 8 h significantly reduced IL-5 ($\log_2$FC −6.5, adjusted p = 0.04), IL-13 (−2.8, $p = 8\times10^{-4}$), IL-4 (−4.1, p = 0.09) and IL-22 (−3.4, p = 0.09), while IFN-γ, TNF and IL-2 were unchanged or increased. This signature closely matches that of knocking down **GATA3**, the master type-2 transcription factor (IL-5 −5.0, IL-13 −4.8; Fig. 2) — yet IMPDH2 knockdown is more selective, because unlike GATA3 loss it does not also suppress FOXP3 and IL-10. A metabolic enzyme knockdown thus reproduces the therapeutic effect of disabling the type-2 lineage's master regulator, without collateral loss of the regulatory compartment.

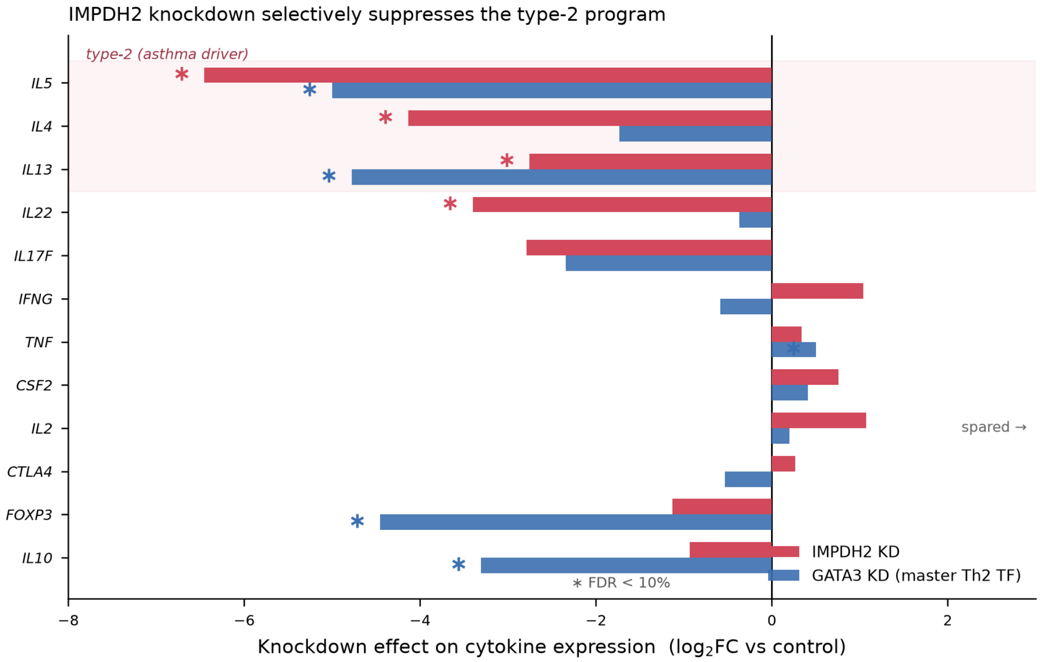

Figure 2. IMPDH2 knockdown selectively suppresses type-2 cytokines. Per-cytokine knockdown effect ($\log_2$FC vs control, 8 h stimulation) for IMPDH2 (red) and GATA3 (blue). Type-2 cytokines (shaded, the asthma-driving axis) are strongly suppressed by both; Th1 (IFN-γ, TNF), proliferation (IL-2) and the regulatory arm (CTLA4) are spared by IMPDH2. GATA3 knockdown additionally suppresses FOXP3/IL-10, making IMPDH2 the more selective node. ∗ FDR < 10%.

3.3 The whole de novo guanine-nucleotide pathway converges on the type-2 program

A single-gene hit could be an artefact. It is not: knocking down the enzymes upstream and around IMPDH2 in the de novo guanine-nucleotide pathway — PPAT, GART, PFAS, PAICS, ATIC and GMPR2 — coherently suppressed the type-2 program, with IMPDH2 (the rate-limiting step) showing the largest effect (Fig. 3). As a set, the eight pathway genes were significantly more type-2-suppressive than all druggable enzymes (mean $\log_2$FC −1.82 vs −0.76; Mann–Whitney $p = 4.8\times10^{-3}$). The constitutive isoform IMPDH1 — not the isoform preferentially inhibited by mycophenolic acid — failed to knock down and showed no effect, consistent with IMPDH2 being the activated-lymphocyte-specific, therapeutically relevant isoform. IMPDH2's knockdown passed all quality-control checks (no off-target or neighbouring-gene effects; cross-donor r = 0.82).

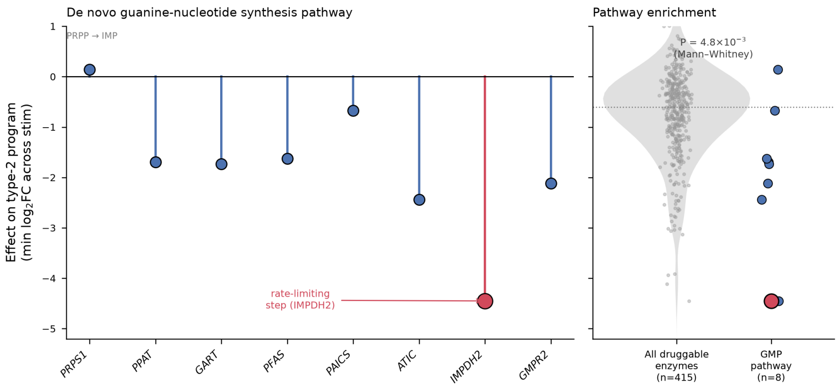

Figure 3. Pathway-level coherence. Left: effect on the type-2 program of knocking down each enzyme of de novo guanine-nucleotide synthesis, in metabolic order; IMPDH2 (red) is the rate-limiting step and strongest suppressor. **Right:** the eight-gene pathway is significantly more type-2-suppressive than the background of all druggable enzymes (n = 415; one-sided Mann–Whitney).

3.4 The nominated axis is the genetic driver of eosinophilic asthma, and the target has a natural-product inhibitor

Which disease does the type-2 axis explain? In the study's Open Targets annotations, the human genetics of **asthma** is dominated by exactly this program: the top risk genes by genetic evidence are IL4R (0.97), IL13 (0.96), IL1RL1/ST2 (0.95), GATA3 (0.95), IL33 (0.93) and TSLP (0.92) — the receptors, alarmins and transcription factors of type-2 inflammation. IMPDH2 itself is not a GWAS hit, which is the point: it is the druggable metabolic checkpoint downstream of the genetically-encoded program, invisible to genetics but revealed by perturbation.

IMPDH2 is the molecular target of **mycophenolic acid (MPA)**, a natural secondary metabolite first isolated from Penicillium in 1893 — among the earliest purified antibiotics. MPA is a potent, reversible inhibitor of IMPDH that is ~5-fold selective for the type II isoform expressed in activated lymphocytes, depleting guanine nucleotides to which proliferating T and B cells are uniquely sensitive [3]. Its morpholinoethyl-ester prodrug, mycophenolate mofetil, is an approved immunosuppressant. Independent of any genomics, mycophenolate mofetil (alongside cyclosporin A and sirolimus) was previously shown to inhibit allergen-driven T-cell proliferation and IL-5 production in peripheral-blood mononuclear cells from atopic asthmatic patients [4] — an experimental result that matches our screen's prediction and provides orthogonal validation.

Table 2. IMPDH2 knockdown, per-cytokine effect at 8 h and 48 h stimulation. Type-2 cytokines are selectively and reproducibly suppressed; Th1 and (at 8 h) proliferation are spared. The IL-2 fall by 48 h reflects the enzyme's known antiproliferative role and informs dose/timing in the pilot.

Cytokine	Program	$\log_2$FC 8h	adj-p 8h	$\log_2$FC 48h	adj-p 48h
IL5	type-2	**−6.45**	0.041	**−3.91**	0.099
IL4	type-2	**−4.14**	0.086	−1.37	0.40
IL13	type-2	**−2.76**	8.4e−4	**−1.70**	2.1e−3
IL22	Th17	**−3.40**	0.095	−3.45	0.36
IFNG	Th1	+1.04	0.19	+0.02	0.99
TNF	Th1	+0.34	0.58	+0.03	0.99
IL2	proliferation	+1.07	0.44	−5.47	0.10
CTLA4	regulatory	+0.26	0.75	+0.75	0.026

4 Optimal pilot experiment

Objective. Test the causal prediction that IMPDH2 inhibition by mycophenolic acid selectively suppresses the type-2 cytokine program in disease-relevant human $CD4^+$ T cells, at concentrations below those that merely block proliferation.

Design. A focused ex vivo dose–response study on primary $CD4^+$ T cells from patients with T2-high (blood-eosinophil-high, elevated FeNO) allergic asthma (target n = 8–12), with non-atopic donors as controls. Isolate $CD4^+$ (and memory Th2/CRTh2$^+$) T cells, stimulate through the TCR (anti-CD3/CD28, or house-dust-mite allergen for the atopic donors, mirroring the reference assay), and treat with mycophenolic acid across a graded concentration range (e.g. 0.1–10 µM) plus vehicle. Because the screen shows the transcriptional type-2 effect precedes the antiproliferative effect, read out at both an **early** (8–24 h) and a **late** (48–72 h) time point.

Primary readouts. (i) Type-2 cytokines IL-5, IL-13, IL-4 (secreted protein by multiplex ELISA/Luminex plus intracellular flow) — prediction: dose-dependent decrease. (ii) Selectivity controls IFN-γ, TNF and regulatory markers FOXP3/CTLA4 — prediction: spared at low/intermediate doses. (iii) A viability/proliferation axis (CellTrace, IL-2, live/dead) to define the therapeutic window. A guanosine add-back arm (restoring the pathway product) confirms on-target action: it should rescue the cytokine effect if IMPDH-mediated.

Mechanistic confirmation and success criterion. In a subset, compare MPA against CRISPRi/siRNA knockdown of IMPDH2 in the same donors; concordant, guanosine-reversible suppression of IL-5/IL-13 closes the loop between genetic prediction and pharmacological agent. Success is a ≥50% reduction in IL-5/IL-13 at an MPA concentration reducing IFN-γ and viability by <20%, reversed by guanosine — justifying an in vivo allergic-airway model and, given mycophenolate mofetil's existing approval, a biomarker-stratified repurposing trial in T2-high asthma.

5 Limitations

Readouts are transcriptional (cytokine mRNA), not secreted protein; the pilot addresses this directly. Several type-2 $\log_2$FC values, though large, have modest FDR owing to the sparsity of cytokine transcripts, but the pathway-level coherence and cross-donor reproducibility mitigate single-gene noise. IMPDH inhibition is antiproliferative at higher exposure — the therapeutic hypothesis rests on a selective transcriptional window that the dose–response pilot is designed to define. Finally, MPA is a natural fungal metabolite rather than a dietary nutraceutical; it is nonetheless a naturally occurring small molecule with an approved prodrug, making translation unusually direct.

References

1. [Dataset] Genome-scale perturb-seq in primary human $CD4^+$ T cells maps context-specific regulators of T cell programs and human immune traits. bioRxiv (2025). doi:10.64898/2025.12.23.696273. Data: CZI Virtual Cell Models (virtualcellmodels.cziscience.com/dataset/genome-scale-tcell-perturb-seq); raw sequencing GEO GSE314342, SRA SRP643211; analysis code github.com/emdann/GWT_perturbseq_analysis_2025.
2. Ochoa D, Hercules A, Carmona M, et al. The next-generation Open Targets Platform: reimagined, redesigned, rebuilt. Nucleic Acids Res. 2023;51(D1):D1353–D1359. doi:10.1093/nar/gkac1046.
3. Allison AC, Eugui EM. Mycophenolate mofetil and its mechanisms of action. Immunopharmacology 2000;47(2–3):85–118. doi:10.1016/S0162-3109(00)00188-0. PMID:10878285.
4. Powell N, Till S, Bungre J, Corrigan C. The immunomodulatory drugs cyclosporin A, mycophenolate mofetil, and sirolimus (rapamycin) inhibit allergen-induced proliferation and IL-5 production by PBMCs from atopic asthmatic patients. J Allergy Clin Immunol. 2001;108(6):915–917. doi:10.1067/mai.2001.119742. PMID:11742267.
5. Regueira TB, Kildegaard KR, Hansen BG, et al. Molecular basis for mycophenolic acid biosynthesis in Penicillium brevicompactum. Appl Environ Microbiol. 2011;77(9):3035–3043. doi:10.1128/AEM.03015-10.